DNA, chromosomes and genes

9A Genetics and evolution · Lesson 4 of 7

Date _____ Class _____

Today I can

- ☐ I can outline how the structure of DNA was discovered and say what each scientist did.
- ☐ I can describe how DNA, chromosomes, genes and genetic information are linked, and give the number of chromosomes in body cells and gametes.

What happens today

1. Do Now: the secret of life
2. Learn: Discovering the structure of DNA · Who did what · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 12 to 13 · Workbook pages 8 to 9

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
DNA	the very long molecule that stores genetic information
double helix	the shape of a DNA molecule: two strands twisted into a spiral
chromosome	a structure in the nucleus made of one long DNA molecule
gene	a section of DNA that carries the instructions for one characteristic
genetic information	the instructions for inherited characteristics
nucleus	the part of a cell that contains the chromosomes
sex chromosomes	the pair of chromosomes that decides sex: XX (female) or XY (male)
gamete	a sex cell (sperm or egg) with 23 chromosomes
zygote	the fertilised egg cell, with 46 chromosomes
X-ray	a type of radiation used to make images, including of molecules
model	a simpler representation of something that scientists test against evidence

Quick Review

the secret of life

28 February 1953, Cambridge. Two men interrupt people eating lunch: 'We have discovered the secret of life!'

1 What do you think they had discovered?

2 Where in a cell is genetic information stored?

3 Guess: how many chromosomes are in one human body cell?

Core 1 of 5

Everyone does these. Answer on the card or under it.

1 Match each scientist to one thing they did to help discover the structure of DNA. Draw a line.

Francis Crick	took very clear X-ray photos of DNA and said the first model was wrong
Rosalind Franklin	tested the model and made corrections to it over several years
James Watson	saw from a photo that DNA was a spiral and helped build the model
Maurice Wilkins	helped to build the double helix model in 1953

2 Answer each question by writing a number.

a) The number of DNA molecules in one chromosome.

Core 2 of 5

Everyone does these. Answer on the card or under it.

2 *continued*

b) The number of Y chromosomes in a body cell from a male.

c) The number of copies of each gene in a body cell from a female.

d) The number of chromosomes in a human gamete.

Core 3 of 5

Everyone does these. Answer on the card or under it.

2 *continued*

e) The number of chromosomes in a zygote.

3 Number these items in order of size. Give the biggest a 1 and the smallest a 4.

____ cell ____ chromosome ____ gene ____ nucleus

Core 4 of 5

Everyone does these. Answer on the card or under it.

4 The photo shows the chromosomes from one human body cell, sorted into pairs.

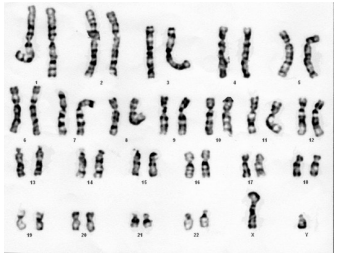


Image: Can H., CC BY 2.0, via Wikimedia Commons

a) State whether the cell is from a male or a female. Give a reason.

Core 5 of 5

Everyone does these. Answer on the card or under it.

4 *continued*

b) How many pairs of chromosomes does the cell contain?

Challenge 1 of 2

When Core is finished. Full sentences.

C1 Draw a labelled diagram to show how DNA, chromosomes, genes, genetic information and the nucleus are linked. Use arrows or nested shapes.

Draw here.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Watson and Crick used data from other scientists to build their model.

a) Suggest how they were able to find other scientists' data in 1953.

b) Explain why scientists publish their work in journals such as Nature, and why it mattered that Franklin and Wilkins published in the same edition.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Where in a cell is DNA found?

2 How many chromosomes are there in a human body cell, and in a human gamete?

3 What is a gene?

Today I can: Goal 1 ☐ Goal 2 ☐